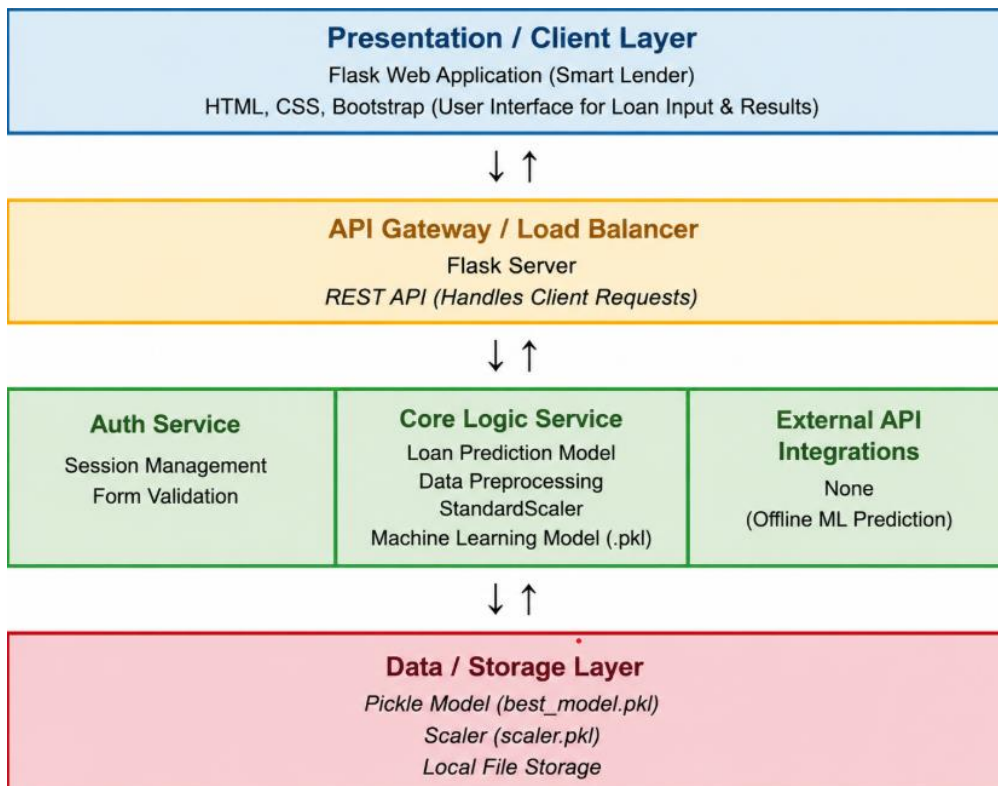


Solution Architecture

Date	30 June 2026
Team ID	XXXXXX
Project Name	Smart Lender
Maximum Marks	5 Marks

Solution Architecture Diagram:

The **Smart Lender** project follows a three-layer architecture consisting of the Presentation Layer, Application Layer, and Data Layer. The presentation layer provides a user-friendly Flask web interface for entering loan details, while the application layer processes the input using a trained machine learning model to predict loan approval. The data layer securely stores the trained model, scaler, and related project files, enabling fast, accurate, and reliable loan prediction.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
 Presentation Layer	Provides the user interface where applicants enter loan details and view the prediction result. Handles form rendering and displays outputs to the user.	Flask, HTML, CSS, Bootstrap, Jinja2
 API Gateway	Receives client requests from the web interface and routes them to the prediction module. Acts as the entry point for all application requests.	Flask (REST API)
 Core Logic Service	Processes the input data, applies preprocessing using the scaler, loads the machine learning model, and predicts whether the loan will be approved or rejected.	Python, NumPy, Scikit-learn, Pickle (best_model.pkl, scaler.pkl)
 Data / Storage Layer	Stores the trained machine learning model and scaler used for prediction. These files are loaded during runtime to generate results.	Pickle (.pkl), Local File Storage